

Business Statistics - MS6107E

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Outline – Part I

1 Linear Regression

- Linear Regression Model
- Correlation
- Linear Regression Model - Numerical Example

2 ANOVA - Analysis of Variance

- Introduction and Assumptions
- One-Way ANOVA: Numerical Example
- Post-Hoc Analysis: Tukey's HSD



Linear Regression

Motivation

- "If we increase our marketing budget by 1 lakh, how much more can we expect in sales?"
- "How does employee training investment impact productivity?"
- "How does advertising expenditure influence sales revenue?"
- "Both marketing spend and number of sales staff together influence total sales."
- "How does product price and household income impact demand"



Linear Regression

Linear Regression Model

The linear regression model in the conditional-expectation form is defined as:

$$\mathbb{E}[Y | X] = \beta_0 + \beta^\top X$$

- Y is the dependent variable
- X is the set of independent variables $X_1, X_2, \dots, X_p$
- p is the number of independent variables
- β is the set of $p+1$ parameters $\beta_0, \beta_1, \dots, \beta_p$
- $\mathbb{E}[Y | X]$ is the prediction made by the model given the set of p independent variables $X_1, X_2, \dots, X_p$

$$\mathbb{E}[Y | X] = \beta_0.1 + \beta_1.X_1 + \beta_2.X_2 \dots \beta_p.X_p$$



Population Model of Linear Regression

Normal Equations

$$Y = \mathbb{E}[Y | X] + \varepsilon$$

$$Y = \beta_0 + \beta^\top X + \varepsilon, \quad \text{where } \mathbb{E}[\varepsilon | X] = 0.$$

$$(\beta_0^*, \beta^*) = \arg \min_{\beta_0 \in \mathbb{R}, \beta \in \mathbb{R}^p} \mathbb{E}[(Y - \beta_0 + \beta^\top X)^2].$$

The minimizer (β_0^*, β^*) satisfies the population normal equations:

$$\mathbb{E}[Y] = \beta_0^* + (\beta^*)^\top \mathbb{E}[X], \quad \mathbb{E}[X(Y - \beta_0^* - (\beta^*)^\top X)] = 0$$

- The expected (average) value of Y in the population equals the predicted value from the regression line, when evaluated at the average of X .
- The “unexplained part” of Y or the error $\varepsilon = Y - \beta_0^* - (\beta^*)^\top X$ is uncorrelated with all the predictors X .



Linear Regression

Ordinary Least Square Estimation

- Given data $\{(y_i, x_{i1}, x_{i2}, \dots, x_{ip})\}_{i=1}^n$, the OLS estimator solves:

$$\hat{\beta} = \arg \min_{\beta} \frac{1}{n} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 \cdot x_{i1} + \dots + \beta_p \cdot x_{ip}))^2$$

■

$$\hat{\beta}_{aug} = (X_{aug}^T X_{aug})^{-1} X_{aug}^T y$$

$$\mathbb{E}[Y | X] = \hat{\beta}_{aug}^T X$$

This is the equation of the line or plane that predicts Y from X by finding the coefficients that minimize the overall squared errors between the predicted and actual values.



Linear Regression

$$\hat{\beta} = \arg \min_{\beta} \frac{1}{n} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 \cdot x_{1i} + \cdots + \beta_p \cdot x_{pi}))^2.$$

$$\hat{\beta} = (X_{\text{aug}}^{\top} X_{\text{aug}})^{-1} X_{\text{aug}}^{\top} y$$

$$X_{\text{aug}} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}_{n \times (p+1)}$$

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1}$$



Linear Regression

Normality Assumption

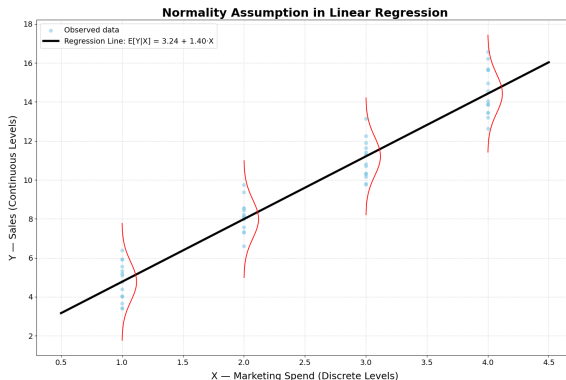


Figure: Linear Regression - Normality Assumption



Linear Regression

Homo-skedasticity Assumption

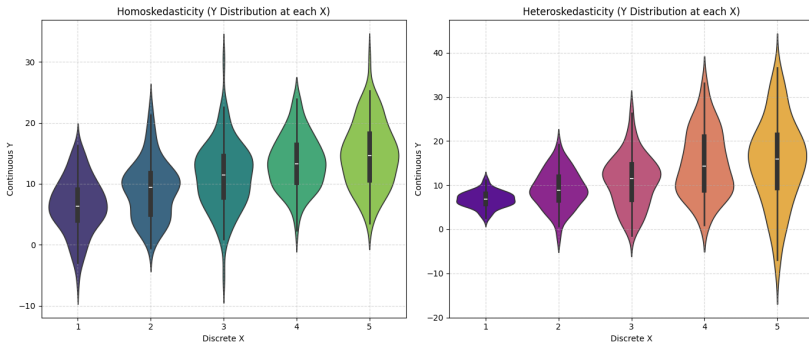


Figure: Linear Regression - Homo-skedasticity Assumption



Linear Regression

Model Sales Revenues as a function of Marketing Spend and Sales Staff

Sample Data (15 observations):

Obs	Marketing Spend (k)	Sales Staff	Sales Revenue (k)
1	10	2	25
2	12	3	30
3	15	4	38
4	8	2	20
5	20	5	50
6	18	4	45
7	25	6	60
8	5	1	12
9	14	3	35
10	22	5	55
11	9	2	22
12	16	4	40
13	13	3	33
14	7	1	15
15	19	5	48



Linear Regression

Model Sales Revenues as a function of Marketing Spend and Sales Staff

Predictor	Estimate	Std. Error	t value	Pr(> t)
(Intercept β_0)	5.00	1.50	3.33	0.005
Marketing Spend β_1	1.50	0.10	15.00	< 0.001
Sales Staff β_2	3.00	0.20	15.00	< 0.001

Table: OLS Regression Results

R-squared = 0.98, **Adjusted R-squared** = 0.97

$$\text{Sales Revenue} = \hat{\beta}_0 + \hat{\beta}_1 \times \text{Marketing Spend} + \hat{\beta}_2 \times \text{Sales Staff}$$



Linear Regression

Correlation Matrix

The correlation coefficient between independent variables should be minimum.

	Marketing Spend	Sales Staff	Sales Revenue
Marketing Spend	1.00	0.98	0.99
Sales Staff	0.98	1.00	0.99
Sales Revenue	0.99	0.99	1.00

Hence, ideally one of the variables need to be removed.



Linear Regression

Model Sales Revenues as a function of Marketing Spend with other Variable Removed

Fitted Model:

$$\text{Sales Revenue} = \hat{\beta}_0 + \hat{\beta}_1 \times \text{Marketing Spend}$$

Predictor	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10.00	1.20	8.33	0.0001
Marketing Spend	2.00	0.12	16.67	< 0.001

Table: OLS Regression Results (Simple Linear Model)

R-squared = 0.95, **Adjusted R-squared** = 0.95



Linear Regression

Correlation Coefficient

The Pearson correlation coefficient r measures the strength and direction of the *linear relationship* between two continuous variables X and Y .

$$r_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

Interpretation:

- $r = 1$: Perfect positive linear relationship
- $r = -1$: Perfect negative linear relationship
- $r = 0$: No linear relationship



Linear Regression

Correlation Coefficient

Pearson Correlation Coefficient:

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]}{\sqrt{\mathbb{E}[(X - \mathbb{E}[X])^2] \mathbb{E}[(Y - \mathbb{E}[Y])^2]}}$$

- $\mathbb{E}[X]$ = expected value (mean) of X
- $\sigma_X = \sqrt{\mathbb{E}[(X - \mathbb{E}[X])^2]}$ = standard deviation of X
- $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$



Linear Regression

Correlation Coefficient & Covariance

Covariance vs. Correlation:

Property	Covariance $\text{Cov}(X, Y)$	Correlation ρ_{XY}
Definition	$\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$	$\frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$
Unit dependence	Depends on units of X and Y	Unit-free (dimensionless)
Range	Unbounded	$[-1, +1]$
Interpretation	Direction of linear relationship	Strength and direction of linear relationship
Scale sensitivity	Changes with variable scaling	Invariant to scaling (standardized)



Linear Regression

Correlation of Linearly Scaled Variables

Correlation of Linearly Scaled Variables

$$\text{Corr}(aX, bY) = \text{sign}(ab)\text{Corr}(X, Y) \quad (1)$$

$$\text{Cov}(aX, bY) = (ab)\text{Cov}(X, Y) \quad (2)$$

- Correlation is scale-invariant — only the sign of a and b matters.
- If $a, b > 0$, correlation remains unchanged.
- If one of a or b is negative, correlation changes sign.
- Covariance is scale variant. Covariance magnitude depends upon the product of a and b .



Linear Regression

Covariance of Normalized Random Variables

Let X and Y be random variables with means μ_X, μ_Y and standard deviations σ_X, σ_Y . The normalized (standardized) forms are:

$$Z_X = \frac{X - \mu_X}{\sigma_X}, \quad Z_Y = \frac{Y - \mu_Y}{\sigma_Y}$$

Then:

$$E[Z_X] = E[Z_Y] = 0, \quad \text{Var}(Z_X) = \text{Var}(Z_Y) = 1$$

$$\text{Cov}(Z_X, Z_Y) = E[(Z_X - E[Z_X])(Z_Y - E[Z_Y])]$$

$$= E[Z_X Z_Y]$$

$$= E \left[\frac{(X - \mu_X)(Y - \mu_Y)}{\sigma_X \sigma_Y} \right]$$

$$= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \rho(X, Y)$$



Linear Regression

Spearman's Correlation Coefficient

Spearman Correlation measures the *monotonic relationship* between two variables using their *ranks*. Given n paired observations (x_i, y_i) :

- 1 Convert each variable to ranks:

$$R(x_i), R(y_i)$$

- 2 Compute the difference in ranks:

$$d_i = R(x_i) - R(y_i)$$

- 3 The Spearman coefficient is given by:

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$



Linear Regression

Spearman's Correlation Coefficient : Interpretation & Difference with Pearson

r_s Value	Interpretation
+1	Perfect positive monotonic relationship
0	No monotonic relationship
-1	Perfect negative monotonic relationship

Aspect	Pearson Correlation	Spearman Correlation
Relationship	Linear relationship	Monotonic relationship (increasing or decreasing)
Data Type	Requires interval or ratio scale	Works Ordinal, interval, or ratio data
Assumption	Assumes normality	Non-parametric
Effect of Outliers	Highly sensitive	Less sensitive

Table: Pearson and Spearman Correlation Coefficient



Correlation Coefficient

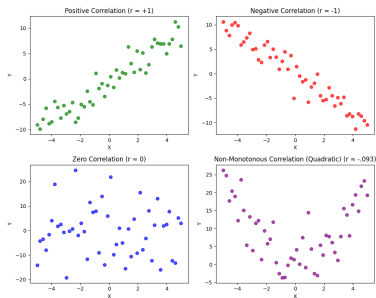


Figure: Correlation Coefficient



Linear Regression

Correlation from Scatter Plot

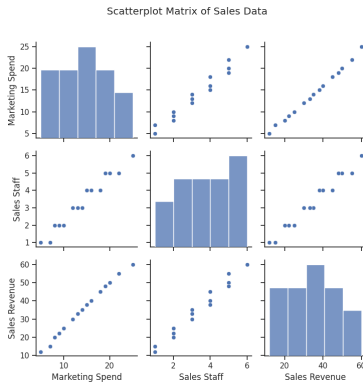


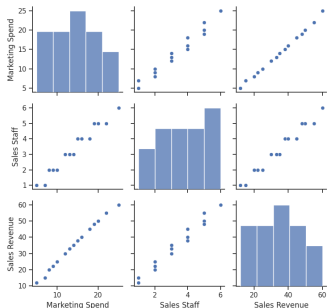
Figure: Scatter Matrix



Linear Regression

Scatter Matrix and Correlation Matrix

Scatterplot Matrix of Sales Data



	MS	SS	SR
Marketing Spend MS	1.00	0.98	0.99
Sales Staff SS	0.98	1.00	0.99
Sales Revenue SR	0.99	0.99	1.00

Figure: Scatter Matrix



Linear Regression

Coefficient of Determination R^2

Definition:

$$R^2 = 1 - \frac{SSR}{SST}$$

where

$$SSR = \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

- y_i : Observed values
- $\hat{y}_i$: Predicted values from the regression model
- $\bar{y}$: Mean of observed y_i
- SSR (Sum of Squared Residuals): Unexplained variation
- SST (Total Sum of Squares): Total variation in Y

Equivalent Form (Correlation-based):

$$R^2 = (\text{Corr}(Y, \hat{Y}))^2$$



Linear Regression

Coefficient of Determination R^2

Definition:

$$R^2 = 1 - \frac{SSR}{SST} = \frac{SST - SSR}{SST} = \frac{\text{Explained Variance}}{\text{Total Variance}}$$

As a Percentage:

$$\text{Percentage of Variance Explained} = R^2 \times 100\%$$

Interpretation:

- R^2 represents the proportion of total variation in Y that is explained by the regression model.
- Expressing it as $R^2 \times 100\%$ gives the percentage of variance in Y explained by the model.
- For example, $R^2 = 0.85$ means the model explains 85% of the variation in Y .



Linear Regression

Linear Regression : Example

Objective : Model Sales Revenues as a function of Advertising Spend AS, Promotions spend PS, Training hours TH

$$\mathbb{E}[Y | X] = \beta_0 + \beta_1.AS + \beta_2.PS + \beta_3.TH$$

Correlation Matrix

	AS	PS	TH
Advertising Spend	1.000000	0.927709	-0.028591
Promotions Spend	0.927709	1.000000	0.005904
Training Hours	-0.028591	0.005904	1.000000



Linear Regression

Linear Regression : Sales = f(A,P,T)

Dep. Variable:	Sales_Revenue	R-squared:	0.652
Model:	OLS	Adj. R-squared:	0.641
Method:	Least Squares	F-statistic:	59.91
Date:	Mon, 27 Oct 2025	Prob (F-statistic):	6.47e-22
Time:	02:58:46	Log-Likelihood:	-428.41
No. Observations:	100	AIC:	864.8
Df Residuals:	96	BIC:	875.2
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P > t	[0.025	0.975]
const	17.9809	4.776	3.765	0.000	8.501	27.460
Advertising_Spend A	0.7550	0.163	4.638	0.000	0.432	1.078
Promotions_Spend P	-0.0500	0.200	-0.250	0.803	-0.447	0.347
Training_Hours T	0.8299	0.126	6.603	0.000	0.580	1.079
Omnibus:	1.663	Durbin-Watson:	2.324			
Prob(Omnibus):	0.435	Jarque-Bera (JB):	1.110			
Skew:	-0.151	Prob(JB):	0.574			
Kurtosis:	3.419	Cond. No.	199.			



Linear Regression

Parameter Estimate $\hat{\beta}$ and t-value

Table: Linear Regression Estimator ($X = X_{aug}$)

Quantity	Mathematical Expression
Estimated Coefficient ($\hat{\beta}_j$)	$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$
Variance of Coefficient	$\text{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$
Estimated Variance	$\widehat{\text{Var}}(\hat{\beta}) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}, \quad \hat{\sigma}^2 = \frac{\sum (y_i - \hat{y}_i)^2}{n - k - 1}$
Standard Error of Coefficient	$SE(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 [(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$
t-statistic	$t_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$



Linear Regression

Parameter Estimate $\hat{\beta}$ and t-value

Model

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2 I)$$

OLS Estimator

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Variance-Covariance Matrix

$$\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$$

Estimated as:

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n - k}$$



Linear Regression

Parameter Estimate $\hat{\beta}$ and t-value

Standard Error of Coefficient

$$SE(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}$$

t-statistic

$$t_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)} \quad \text{with } (n - k - 1) \text{ d.f.}$$



Linear Regression

Parameter Estimate $\hat{\beta}$ and t-value

Model

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon_i$$

Hypotheses for a Coefficient β_j

$$H_0 : \beta_j = 0 \quad (\text{no effect})$$

$$H_1 : \beta_j \neq 0 \quad (\text{has effect})$$

t-statistic

$$t_j = \frac{\hat{\beta}_j - 0}{SE(\hat{\beta}_j)}$$

$$t_j \sim t_{(n-k-1)} \quad \text{under } H_0$$



Linear Regression

Parameter Estimate $\hat{\beta}$ and t-value

Decision Rule

Reject H_0 if:

$$|t_j| > t_{\alpha/2, (n-k-1)} \quad \text{or} \quad p\text{-value} < \alpha$$

Interpretation

- Reject H_0 : variable x_j significantly affects y .
- Fail to reject H_0 : no significant effect of x_j .



Linear Regression

Simple Linear Regression - Model

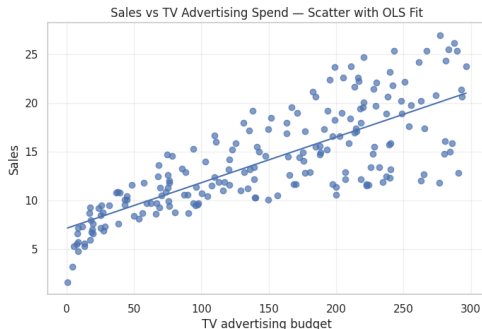


Figure: Sales $y \sim$ TV Advertising Budget x_1 : Data (200*2)



Model Summary and Diagnostics

Table: Summary of Linear Regression Model

Dataset Information	
Number of Observations	200
Number of Variables	4 (TV, Radio, Newspaper, Sales)
Sample Rows	
1	TV = 230.1, Radio = 37.8, Newspaper = 69.2, Sales = 22.1
2	TV = 44.5, Radio = 39.3, Newspaper = 45.1, Sales = 10.4
3	TV = 17.2, Radio = 45.9, Newspaper = 69.3, Sales = 9.3
Fitted Model	$\hat{y} = 7.1318 + 0.0469x$
Model Performance	
R^2_{Train}	0.5907
$\text{MSE}_{\text{Train}}$	10.9612
R^2_{Test}	0.6606
MSE_{Test}	9.1793
Residual Diagnostics	
Durbin-Watson Statistic	1.9985
Shapiro-Wilk (Test Residuals)	$W = 0.9635, p = 0.1240$



Linear Regression

Simple Linear Regression - Training/Testing Split

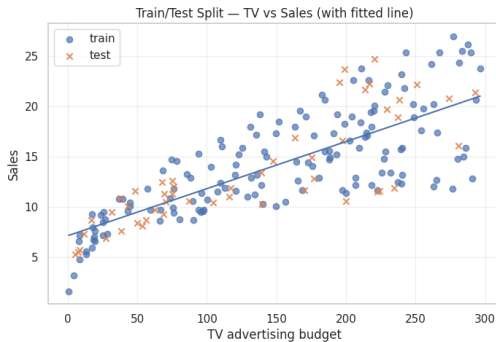


Figure: Training Data : Testing Data - 75:25



Linear Regression

Simple Linear Regression - Training Error

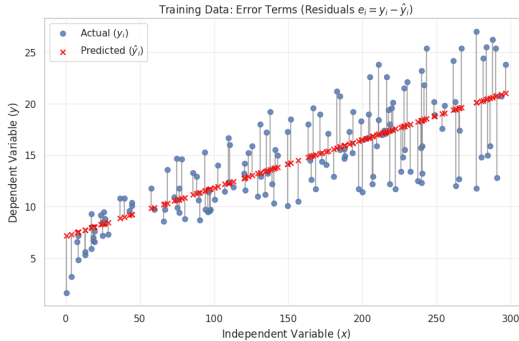


Figure: Error in Linear Regression Model



Linear Regression

Simple Linear Regression - Training Error $\sim (0, \sigma^2)$

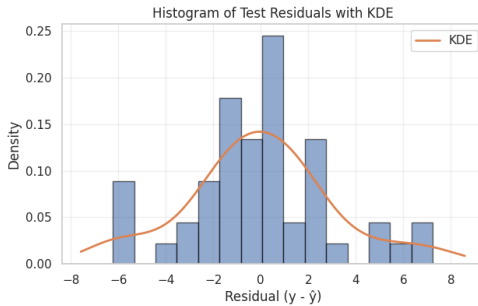


Figure: Residual Analysis



Linear Regression

Simple Linear Regression - Training Error and Predicted Dependent Variable

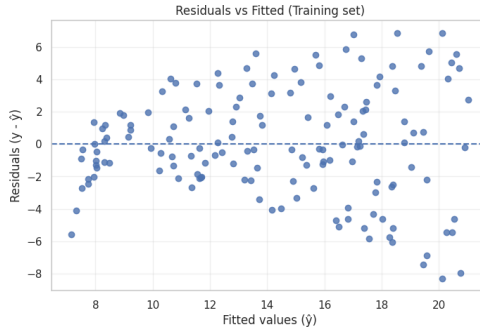


Figure: Residual Analysis



Linear Regression

Simple Linear Regression - Testing Error and Predicted Dependent Variable

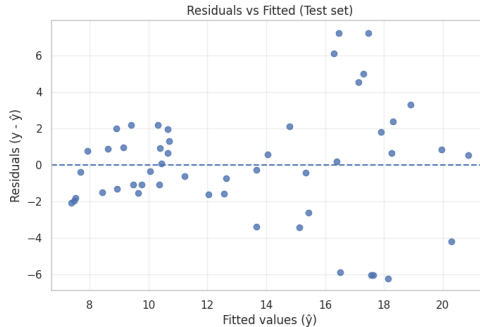
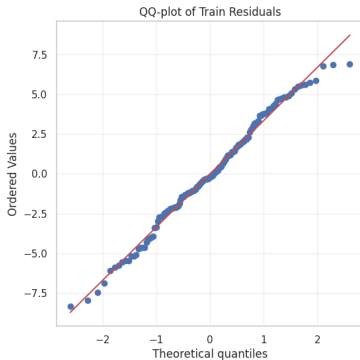


Figure: Residual Analysis

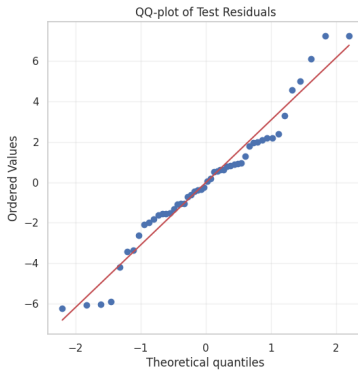


Linear Regression

Simple Linear Regression - QQ Plot Training Set



Simple Linear Regression - QQ Plot Testing Set



Simple Linear Regression

One Dependent and One Independent Variable

Model Equation:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, 2, \dots, n$$

Slope (OLS Estimator):

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\text{Cov}(x, y)}{\text{Var}(x)} = r \frac{s_y}{s_x}$$

where r is the Pearson correlation coefficient, and s_x, s_y are the sample standard deviations of x and y .

Intercept:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$



Simple Linear Regression

One Dependent and One Independent Variable

Coefficient of Determination:

$$R^2 = \frac{SS_{\text{reg}}}{SS_{\text{tot}}} = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = r^2$$

$$\text{where } SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2, \quad SS_{\text{res}} = \sum_i (y_i - \hat{y}_i)^2, \quad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Unbiased Estimator of Error Variance:

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



Simple Linear Regression

One Dependent and One Independent Variable

Key Identities (for OLS with intercept):

$$\sum_{i=1}^n (y_i - \hat{y}_i) = 0, \quad \sum_{i=1}^n \hat{y}_i (y_i - \hat{y}_i) = 0, \quad \sum_{i=1}^n x_i (y_i - \hat{y}_i) = 0$$

No.	Expression	Interpretation
1	$\sum_{i=1}^n (y_i - \hat{y}_i) = 0$	The residuals sum to zero, ensuring that the regression line passes through the mean point $(\bar{x}, \bar{y})$.
2	$\sum_{i=1}^n x_i (y_i - \hat{y}_i) = 0$	The residuals are uncorrelated (orthogonal) with the independent variable x . This guarantees the best linear unbiased estimate (BLUE) under Gauss–Markov assumptions.
3	$\sum_{i=1}^n \hat{y}_i (y_i - \hat{y}_i) = 0$	The residuals are also uncorrelated with the fitted values $\hat{y}_i$. The explained $\hat{y}_i$ and unexplained components $y_i - \hat{y}_i$ of y are orthogonal.



Linear Regression

SS_{tot} - Measure of Variance of Dependent Variable (Training Set)

$$SS_{tot} = \sum_i (y_i - \bar{y})^2 = 4017.046$$

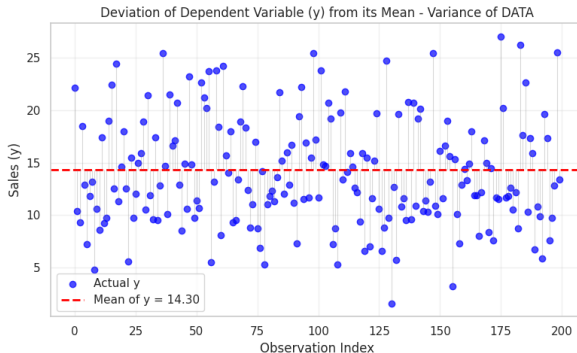


Figure: Variance of Dependent Variable



Linear Regression

Sum of Squares - Linear Regression

$$\text{Regression Sum of Squares (SSR)} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$\text{Total Sum of Squares (SST)} = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$\text{Error Sum of Squares (SSE)} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\text{SST} = \text{SSR} + \text{SSE}$$



Linear Regression

SS_{tot} - Measure of Variance of Model Predicted Dependent Variable (Training Set)

$$SS_{reg} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = 2372.8716$$

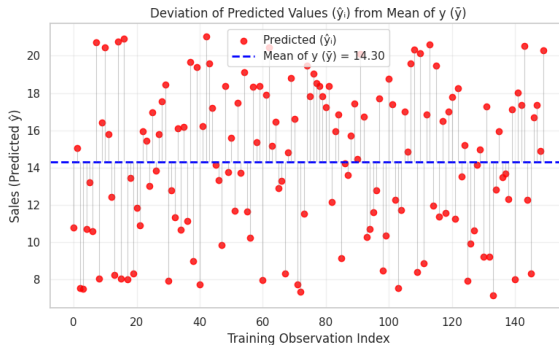


Figure: Variance of Dependent Variable Explained by Model



Linear Regression

SS_{error} - Measure of Variance of Model Error (Training Set)

$$R^2 = \frac{SS_{reg}}{SS_{tot}} = \frac{2372.8716}{4017.0467} = .5907$$

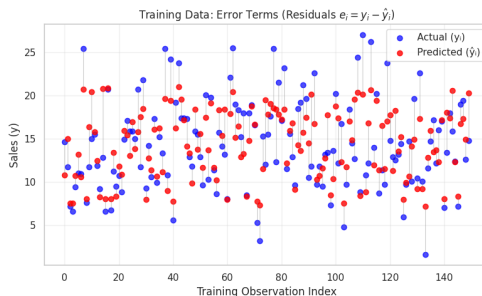


Figure: Variance of the Error in Model $SS_{error} = 1644.1751$



Multiple Linear Regression $R^2 = \rho^2$

Sales $\sim$ TV, Radio, Newspaper

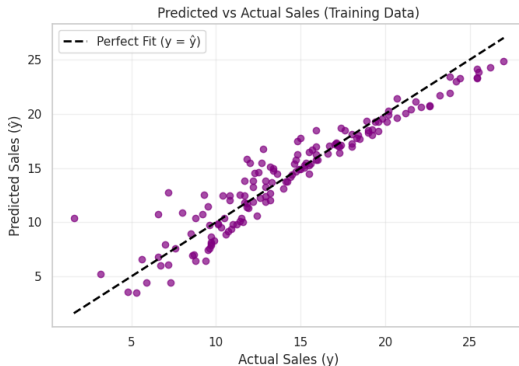


Figure: Pearson Correlation Coefficient $\rho = .9466$ and $R^2 = .8966$



Hypothesis for Multiple Population Means

Analysis of Variance (ANOVA)

ANOVA

Analysis of Variance (ANOVA) is used when:

- The independent variable (X) is **categorical** with k groups.
- The dependent variable (Y) is **continuous**.

ANOVA Model

For a one-way ANOVA:

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim N(0, \sigma^2)$$

where:

- μ is the overall mean,
- τ_i is the effect of the i^{th} group,
- ε_{ij} is the random error for observation j in group i .



Hypothesis for Multiple Population Means

Analysis of Variance (ANOVA)

Hypothesis Testing

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \quad (\text{all group means equal})$$

$$H_A : \text{At least one mean differs}$$

Test Statistic

$$F = \frac{\text{Between-Group Variance (MS}_B\text{)}}{\text{Within-Group Variance (MS}_W\text{)}} = \frac{SS_B/(k-1)}{SS_W/(N-k)}$$

If $F > F_{\alpha, k-1, N-k}$, reject H_0 .

- ANOVA generalizes the two-sample t -test to more than two groups.
- A significant F indicates that at least one group mean differs.



Hypothesis for Multiple Population Means

Analysis of Variance (ANOVA)

- Independence - Observations are independent both within and between groups.
- Normality - The residuals (errors) in each group are normally distributed.
- Homogeneity of variances - The variances across the groups are equal.
- ANOVA assumes each group's data are drawn from a normally distributed population with population mean μ_i , where $\mu_i = \mu + \tau_i$, μ is the overall mean, τ_i is the effect of group i .
- ANOVA is robust to modest violations of normality, especially when group sizes are equal. Hence, we assume group size as equal.



Hypothesis for Multiple Population Means

Analysis of Variance (ANOVA)

A public health researcher is studying whether three types of physical therapy differ in their effectiveness at improving **post-surgery recovery speed** (measured as the number of **days needed for a patient to regain normal mobility**).

Three types of therapy are randomly assigned to $n = 30$ knee-surgery patients (10 per group):

Group	Therapy Type	Description
1	Conventional Therapy (CT)	Standard hospital-guided physiotherapy.
2	Hydrotherapy (HT)	Exercises performed in a temperature-controlled pool.
3	Robot-Assisted Therapy (RT)	Sessions using robotic gait-assist devices.

The dependent variable is **Recovery Days** (continuous variable). Lower recovery days imply faster recovery and hence a more effective treatment.



Hypothesis for Multiple Population Means

Analysis of Variance (ANOVA)

Patient Group	Sample Recovery Days (10 patients each)
CT	33, 37, 41, 35, 39, 36, 38, 34, 40, 37
HT	28, 30, 27, 31, 29, 33, 30, 32, 28, 29
RT	24, 25, 23, 26, 27, 22, 25, 24, 26, 23

Is there a statistically significant difference in the mean recovery time among the three therapy methods?

- **Independent Variable:** Therapy Type (categorical, 3 levels: CT, HT, RT)
- **Dependent Variable:** Recovery Days (continuous)
- **Sample Size:** $n = 30$ (10 per group)



Hypothesis for Multiple Population Means

Analysis of Variance (ANOVA)

$$H_0 : \mu_{CT} = \mu_{HT} = \mu_{RT}$$

H_1 : At least one group mean is different.

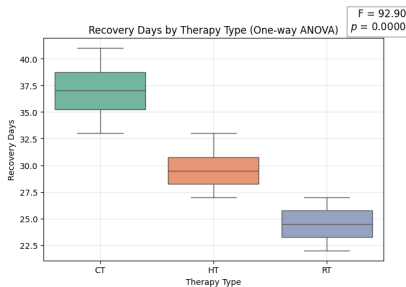


Figure: ANOVA- One way Test



Hypothesis for Multiple Population Means

Analysis of Variance (ANOVA)

1. Total Degrees of Freedom:

$$df_{\text{Total}} = n - 1$$

Measures how all observations vary around the grand mean.

2. Between-Group Degrees of Freedom:

$$df_{\text{Between}} = k - 1$$

There are k group means, but only $k - 1$ can vary independently.

3. Within-Group Degrees of Freedom:

$$df_{\text{Within}} = \sum_{i=1}^k (n_i - 1) = n - k$$

Each group contributes $(n_i - 1)$ degrees of freedom around its own mean.



Hypothesis for Multiple Population Means

Analysis of Variance (ANOVA)

Data summary: $n = 30$ observations, $k = 3$ groups (CT, HT, RT).
Grand mean: $\bar{y} = 30.4$.

Source	SS	df	MS = SS/df	F = MS _B /MS _W	p-value
Between Groups (Regression)	788.6000	2	394.3000	2*92.8979	2*0.001
Within Groups (Error / Residual)	114.6000	27	4.2444		
Total	903.2000	29	—		

$$SS_{\text{Total}} = \sum_{i=1}^n (y_i - \bar{y})^2 = 903.2,$$

$$SS_{\text{Between}} = \sum_{g=1}^k n_g (\bar{y}_g - \bar{y})^2 = 788.6,$$

$$SS_{\text{Within}} = \sum_{g=1}^k \sum_{i \in g} (y_i - \bar{y}_g)^2 = 114.6.$$



Hypothesis for Multiple Population Means

Analysis of Variance (ANOVA)

$$df_{\text{Between}} = k - 1 = 2, \quad df_{\text{Within}} = n - k = 27,$$

$$MS_{\text{Between}} = \frac{788.6}{2} = 394.3000$$

$$MS_{\text{Within}} = \frac{114.6}{27} \approx 4.2444$$

$$F = \frac{MS_{\text{Between}}}{MS_{\text{Within}}} = \frac{394.3000}{4.2444} \approx 92.8979$$

Conclusion: The observed $F \approx 92.90$ with $df = (2, 27)$ gives a p -value $\ll 0.001$. Thus we reject H_0 and conclude at usual significance levels that *not all group means are equal* (i.e., at least one therapy differs in mean recovery days).



Hypothesis for Multiple Population Means

Analysis of Variance (ANOVA)

After finding a significant F -statistic in ANOVA, Tukey's Honest Significance Test (HSD) is used to determine *which specific group means* differ significantly, while **controlling the familywise error rate (FWER)**.

Test statistic:

$$q = \frac{|\bar{Y}_i - \bar{Y}_j|}{\sqrt{\frac{MS_{Within}}{n}}}$$

where:

- $\bar{Y}_i, \bar{Y}_j$ = sample means of groups i and j
- MSE = Mean Square Error (pooled within-group variance from ANOVA)
- n = number of observations per group (assumed equal)



Hypothesis for Multiple Population Means

Analysis of Variance (ANOVA)

Decision rule:

Reject H_0 if $q > q_{\alpha, k, df_{within}}$

where $q_{\alpha, k, df_{within}}$ is the critical value from the *studentized range distribution*, α is the level of significance, k is the number of groups and df_{within} or $n-k$ is the degrees of freedom of within variance.

Interpretation:

- If $q > q_{critical}$, the two group means differ significantly.
- Tukey's HSD adjusts for multiple comparisons, keeping the overall α at the desired level (e.g., 0.05).



Hypothesis for Multiple Population Means

Analysis of Variance (ANOVA)

Why not run multiple T-Tests instead of Tukey's HSD?

- For k groups, $m = \binom{K}{2}$ combinations of T-Tests has to be conducted.
For eg: Groups A,B,C - T-Tests to be conducted for Groups A & B, B & C, C & A.
- The family wise error rate FWER is the probability that at least one of these tests incorrectly rejects a true null hypothesis or the probability of making atleast one type 1 error.
- $\text{FWER} = 1 - (1 - \alpha)^m$, where m is the total number of T-tests to be conducted. For $\alpha = .05$ and $m = 10$, $\text{FWER} = .401$ or there is at-least 40% of chance of making one rejection of null hypotheses when it is indeed true.
- Effectively, using T-test instead of Tukey's HSD might result in rejection of the null hypothesis, when it is indeed true.



References I

